

Second day of class - 8/28

Some elementary number theory

In elementary number theory, the main objects are integers. We denote the set of integers by $\mathbb{Z}$.

Definition. If $a, b \in \mathbb{Z}$ we say that $a|b$ (a divides b) if $b = ka$ for some $k \in \mathbb{Z}$. This is either saying that $a = b = 0$ or that b/a is an integer.

Examples: $3|18$ because $18 = 6 \cdot 3$. (Here $a = 3$, $b = 18$ and $k = 6$.) Also, $7|999999$ (because $7 \cdot 142857 = 999999$).

Theorem. Suppose that $a, b, c, r, s \in \mathbb{Z}$.

- (1) If $a|b$ and $a|c$, then $a|rb + sc$.
- (2) If $a|b$ and $a|c$, then $a|bc$ (and in fact $a^2|bc$).
- (3) If $a|b$ and $b|a$, then $a = \pm b$.

Application: The only integer solutions to $x^3 + y^3 = 1$ are $x = 1$ and $y = 0$, and $x = 0$ and $y = 1$.

We can factor $x^3 + y^3$ as $(x + y)(x^2 - xy + y^2)$, so $(x + y)(x^2 - xy + y^2) = 1$. Thus, $x + y$ and $x^2 - xy + y^2$ must both be equal to 1 or both be equal to -1 .

Since $x^2 - xy + y^2 = x^2 - xy + \frac{1}{4}y^2 + \frac{3}{4}y^2 = (x - \frac{y}{2})^2 + \frac{3}{4}y^2 \geq 0$, the only possibility is for $x^2 - xy + y^2 = 1$ and $x + y = 1$. Taking the second equation and writing $y = 1 - x$ and plugging this into the first equation gives

$$x^2 - x(1 - x) + (1 - x)^2 = 1$$

and so $x^2 - x + x^2 + 1 - 2x + x^2 = 1$, or $3x^2 - 3x + 1 = 1$. Hence $3x^2 - 3x = 3x(x - 1) = 0$. Thus, $x = 0$ or $x = 1$. If $x = 0$, then $y = 1 - x = 1$ and if $x = 1$, then $y = 1 - x = 0$.

GCDs

Definition. If $a, b \in \mathbb{Z}$ are not both zero, we define the greatest common divisor of a and b , $\gcd(a, b)$ to be the largest integer that divides both a and b .

Example: We have $\gcd(75, 105) = 15$. We have $\gcd(311, 106) = 1$.

Q: How do you compute $\gcd(a, b)$?

A: The Euclidean algorithm. Divide b by a and get remainder r_1 ,

$$b = ka + r_1.$$

Now, divide a by r_1 and get a remainder of r_2

$$a = t_1 r_1 + r_2.$$

Now divide r_1 by r_2 and get a remainder r_3 .

$$r_1 = t_2 r_2 + r_3.$$

Each time you do this, the remainder gets smaller. Keep doing this until you get to a remainder of zero. The GCD is the second to last remainder (the one before you hit zero).

Example with 311 and 106.

$$311 = 2 \cdot 106 + 99$$

$$106 = 99 + 7$$

$$99 = 7 \cdot 14 + 1$$

$$7 = 7 \cdot 1 + 0.$$

We have $r_1 = 99$, $r_2 = 7$, $r_3 = 1$ and $r_4 = 0$. The second-to-last remainder is 1.

Theorem. *If $a, b \in \mathbb{Z}$ are not both zero, there are integers x and y so that $ax + by = \gcd(a, b)$.*

You can find x and y by running the Euclidean algorithm backward.

$$\begin{aligned} 1 &= 99 - 14 \cdot 7 = 99 - 14 \cdot (106 - 99) = 15 \cdot 99 - 14 \cdot 106 \\ &= 15 \cdot (311 - 2 \cdot 106) - 14 \cdot 106 \\ &= 15 \cdot 311 - 44 \cdot 106. \end{aligned}$$

This is helpful when doing *modular arithmetic*.

Modular arithmetic

Definition. *If $a, b, m \in \mathbb{Z}$ and $m \geq 1$ we say that $a \equiv b \pmod{m}$ if $m \mid a - b$. This is the same as saying that a and b have the same remainder when divided by m . We say that a is congruent to b modulo m .*

Ex: We have $104 \equiv 53 \pmod{17}$ because 17 divides $104 - 53 = 51 = 3 \cdot 17$. Alternatively

$$104 = 6 \cdot 17 + 2$$

$$53 = 3 \cdot 17 + 2.$$

Properties of modular arithmetic:

1. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$.
2. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $ac \equiv bd \pmod{m}$.
3. For any $a \in \mathbb{Z}$, there is a unique r , $0 \leq r \leq m - 1$ so that $a \equiv r \pmod{m}$.
4. If $ca \equiv cb \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b \pmod{m}$.

Note that this is false if $\gcd(c, m) > 1$. For example, $2 \cdot 2 = 4$ and $2 \cdot 5 = 10$. We have that $4 \equiv 10 \pmod{6}$. However, $2 \not\equiv 5 \pmod{6}$.

5. If $ca \equiv cb \pmod{m}$, then $a \equiv b \pmod{\frac{m}{\gcd(c, m)}}$.

Perspective: Given a positive integer m , we will let $\mathbb{Z}/m\mathbb{Z}$ be the set $\{0, 1, 2, \dots, m-1\}$. Any integer is congruent to one of these numbers mod m . For $a, b \in \mathbb{Z}/m\mathbb{Z}$, we define $a + b$ to be the remainder when $a + b$ is divided by m and $a \cdot b$ to be the remainder when ab is divided by m . This makes it so we can add and multiply elements of $\mathbb{Z}/m\mathbb{Z}$ and stay in this set. (This is an important idea.)

6. If $\gcd(a, m) = 1$, then there is an integer x so that $ax \equiv 1 \pmod{m}$.

Proof: Since $\gcd(a, m) = 1$, there are integers x and y so that $ax + my = 1$. Thus, $ax - 1 = -my$ and so $m \mid ax - 1$ and $ax \equiv 1 \pmod{m}$.

Notes: This x is the multiplicative inverse of a modulo m . For all practical purposes, this can be thought of as the reciprocal of a modulo m .

Ex: The multiplicative inverse of 5 modulo 7 is 3. This is because $3 \cdot 5 = 15 \equiv 1 \pmod{7}$.

(So the congruence $3x \equiv 1 \pmod{7}$ is equivalent to $x \equiv 5 \pmod{7}$.)

Application: If $a \in \mathbb{Z}$, then there are integers b and c so that $a = b^2 - c^2$ if and only if $a \not\equiv 2 \pmod{4}$.

Proof:

Assume that $a = b^2 - c^2$.

Note that if $x \in \mathbb{Z}$, then x is congruent to 0, 1, 2 or 3 mod 4. Thus, x^2 is congruent to one of $0^2 = 0$, $1^2 = 1$, $2^2 = 4 \equiv 0 \pmod{4}$ or $3^2 = 9 \equiv 1 \pmod{4}$, so any square is congruent to 0 or 1 mod 4.

So a difference of two squares must be congruent to either $0 - 0$, $1 - 0$, $0 - 1$ or $1 - 1$. These are congruent to 0, 1, 3 and 0 respectively. We cannot get a remainder of 2 and so $a \not\equiv 2 \pmod{4}$.

Assume that $a \not\equiv 2 \pmod{4}$.

There are two cases. If a is odd, say $a = 2k + 1$, then we can write $a = (k + 1)^2 - k^2$. If a is even, then since $a \not\equiv 2 \pmod{4}$, we must have $a \equiv 0 \pmod{4}$, and so $a = 4k$. Then $a = (k + 1)^2 - (k - 1)^2$. This proves that a is the difference of two squares.

QED

Another example: Solutions to $x^2 - 5y^2 = 2$. There aren't any. (What about $x^2 - 5y^2 = 1$?).

Primes and the fundamental theorem of arithmetic

Definition. A prime number is a number $p > 1$ with the property that its only divisors are 1 and itself.

Ex: 2, 3, 5, 101, 163, $9 \cdot 10^9 + 1$, 111111111111111111,

Something special happens for primes. If $a \not\equiv 0 \pmod{p}$, then $p \nmid a$. This means that $\gcd(a, p) = 1$, because any common divisor must divide p . It follows that there is $b \in \mathbb{Z}$ so that $ab \equiv 1 \pmod{p}$. This b is the “reciprocal” of a modulo p . Another way of saying this is that if p is prime, you can divide by anything nonzero in $\mathbb{Z}/p\mathbb{Z}$.

Theorem. If $n \in \mathbb{Z}$ and $n \geq 1$, then n can be written uniquely as a product of powers of prime numbers.

Ex: $2808 = 2^3 \cdot 3^3 \cdot 13$.

Q: How many divisors does 2808 have?

A: Any divisor must have the form $2^a \cdot 3^b \cdot 13^c$ and $a \leq 3$, $b \leq 3$ and $c \leq 1$. There are 4 choices for a , 4 choices for b and 2 choices for c for a total of $4 \cdot 4 \cdot 2 = 32$.

Application: $\log_2(3)$ is irrational.

Application: The only integer solution to $y^2 = x^3 + x$ is $x = y = 0$.

We have $y^2 = x(x^2 + 1)$. If $x \neq 0$, then since $y^2 \geq 0$, we must have $x \geq 1$.

Claim: $\gcd(x, x^2 + 1) = 1$. If $d|x$ and $d|x^2 + 1$, then $d|x^2$, and so $d|(x^2 + 1) - x^2 = 1$.

Now assume that $y^2 = x(x^2 + 1)$ with $y > 0$. Then we have two relatively prime numbers whose product is a square.

Let $x = \prod_{i=1}^r p_i^{k_i}$ and $x^2 + 1 = \prod_{j=1}^s q_j^{l_j}$ be the prime factorizations of the factors. Then because $\gcd(x, x^2 + 1) = 1$, there are no prime factors shared between x and $x^2 + 1$.

Now

$$x(x^2 + 1) = \left(\prod_{i=1}^r p_i^{k_i} \right) \left(\prod_{j=1}^s q_j^{l_j} \right)$$

is a perfect square. But there is no overlap in the prime factors. This means that k_i and l_j must be even for all i and j . This makes it so that $x = z^2$ and $x^2 + 1 = w^2$ for some integers z and w . Using $x^2 + 1 = w^2$, we have $x^2 - w^2 = -1$ and so $(x - w)(x + w) = -1$. This means that either $x - w = 1$ and $x + w = -1$ or $x - w = -1$ and $x + w = 1$. Adding gives that $2x = 0$ and so $x = 0$.

Hence $x = 0$ is the only solution.

Note: This method doesn't always work. If $y^2 = x^3 - 2x$, then we can have $\gcd(x, x^2 - 2)$ equal to 1 or 2. Apart from $x = -1$, we can then have $x = w^2$ and $x^2 - 2 = z^2$, which gives the solution $x = y = 0$, or $x = 2w^2$ and $x^2 - 2 = 2z^2$. There are two solutions here: $x = y = 2$ and $x = 338$, $y = 6214$. (Proving there are only two solutions is fairly difficult.)

Fields

A field F is a collection of numbers that can be added and multiplied. This collection of numbers must have the properties that if

$0 \in F$,

if $a, b \in F$, then $a + b \in F$ and $a \cdot b \in F$

addition and multiplication are commutative and associative, and multiplication distributes over addition

if $a \cdot b = 0$, then $a = 0$ or $b = 0$

if $a \in F$ and $a \neq 0$, then there is a $b \in F$ so that $ab = 1$. (Another way of saying this is $1/b \in F$.)

Examples of fields:

$\mathbb{Q}$, the set of rational numbers is a field

$\mathbb{R}$, the set of real numbers is a field

$\mathbb{C}$, the set of complex numbers is a field

If p is prime, $\mathbb{Z}/p\mathbb{Z}$ is a field (and we often write it as $\mathbb{F}_p$)

Example of non-fields:

The set of 2×2 matrices is not a field. You can have two different nonzero matrices that multiply to be zero.

If n is not prime, then $\mathbb{Z}/n\mathbb{Z}$ is not a field. This is because $n = a \cdot b$ for $1 < a < n$ and $1 < b < n$. Then a and b are not zero in $\mathbb{Z}/n\mathbb{Z}$, but $a \cdot b = n \equiv 0 \pmod{n}$ is zero. This fact is the key to an algorithm that uses elliptic curves to factor.